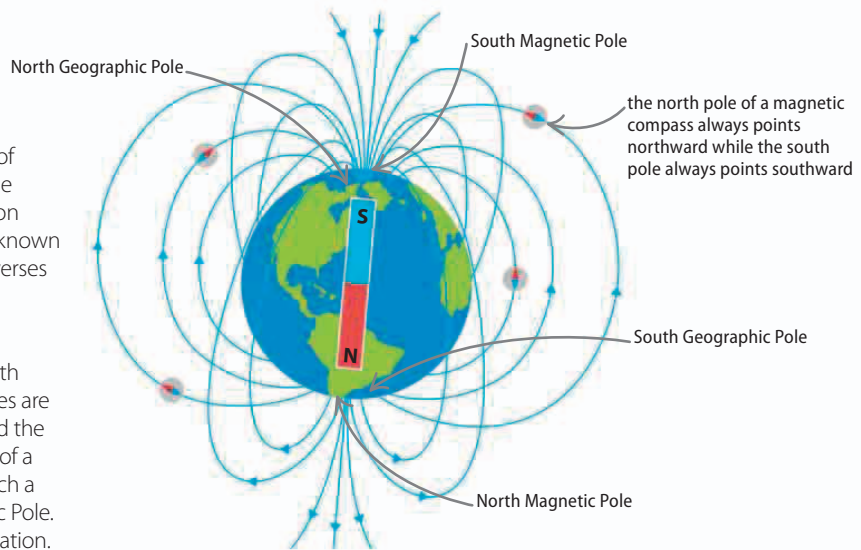


## Earth's magnetic field

The Earth can be thought of as one big powerful magnet whose magnetic field, called the magnetosphere, stretches tens of thousands of kilometers out into space. The planet's magnetism is caused by the motion of liquid metals in its outer core. For an unknown reason, the direction of the Earth's field reverses suddenly, about once every million years.

### ▷ Magnetic Earth

The magnetic pole at Earth's north is a south pole, because the north poles of compasses are attracted by it. Confusingly, it is often called the South Magnetic Pole. There is a difference of a few degrees between the direction in which a compass points and the North Geographic Pole. This difference is called the angle of declination.

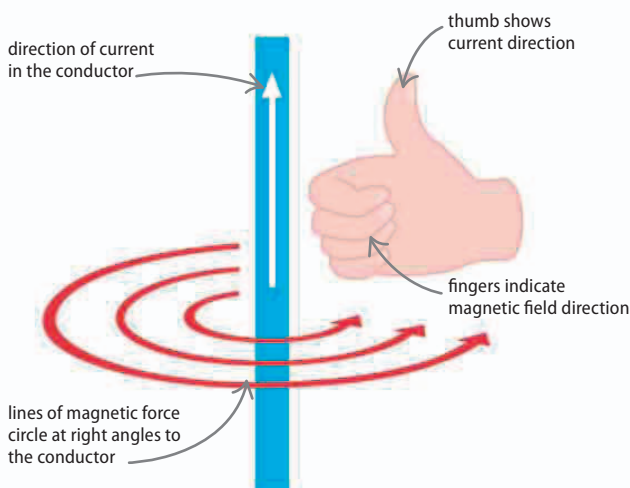


## Electromagnet

Magnets are not the only source of magnetic fields. An electric current flowing through a conductor produces a circular magnetic field at right angles to the conductor. The current creates an electromagnet—a device that is extremely useful since its magnetism can be controlled and switched on and off. The poles of an electromagnet will be reversed if the direction of the current is reversed.

### ▽ Field direction

The direction of the magnetic field can be remembered by making a loose fist with your fingers of your right hand as if grasping the conductor. Sticking your thumb up in the direction of the current, your fingers follow a curving path in the direction of the magnetic field.



### ▽ Solenoid

A solenoid is a common form of electromagnet. It consists of a coil of wire through which an electric current is passed to produce a magnetic field. The soft iron core in the middle of this solenoid helps produce a stronger magnetic field and does not retain its magnetism after the current is switched off.

